

COS30045 Project Stand Up 1

TOPIC: HOSPITALISED INJURIES FROM ROAD CRASHES IN
AUSTRALIA

BY: HELEN AND AKHILA

Project Overview

- **Topic:** Road crash hospitalised injuries in Australia (2011-2021)
- **Focus:** First Nations vs Non-Indigenous disparities
- **Key question our visualisation will answer:**
 1. How have trends changed over 11 years?
 2. Which age/road user/remoteness groups are highest risk?
 3. Where are the largest equity gaps?

Project Setup & Data Identification

1. Project Design Book – FULL DRAFT finished

- Introduction, background, motivation, purpose
- Data source & governance documented

2. Data identified & collected

- BITRE/ First-Nations-hospitalised-injuries-2023.xlsx

3. Data dictionary fully reviewed

4. KNIME workflow built

- Data import, cleaning, reshaping, unpivot

5. GitHub Classroom team created & shared

- <https://github.com/Sixxx-cake/Data-Visualisation---Project>

Data Source

- Source: AIHW National Hospital Morbidity Database
- Published by: BITRE (Bureau of Infrastructure, Transport, Regional Development, Communications and the Arts)
- File used: First-Nations-hospitalised-injuries-2023.xlsx
- Period: 2011-2021 (11 years)
- Privacy: Fully de-identified – only group totals, no individual records

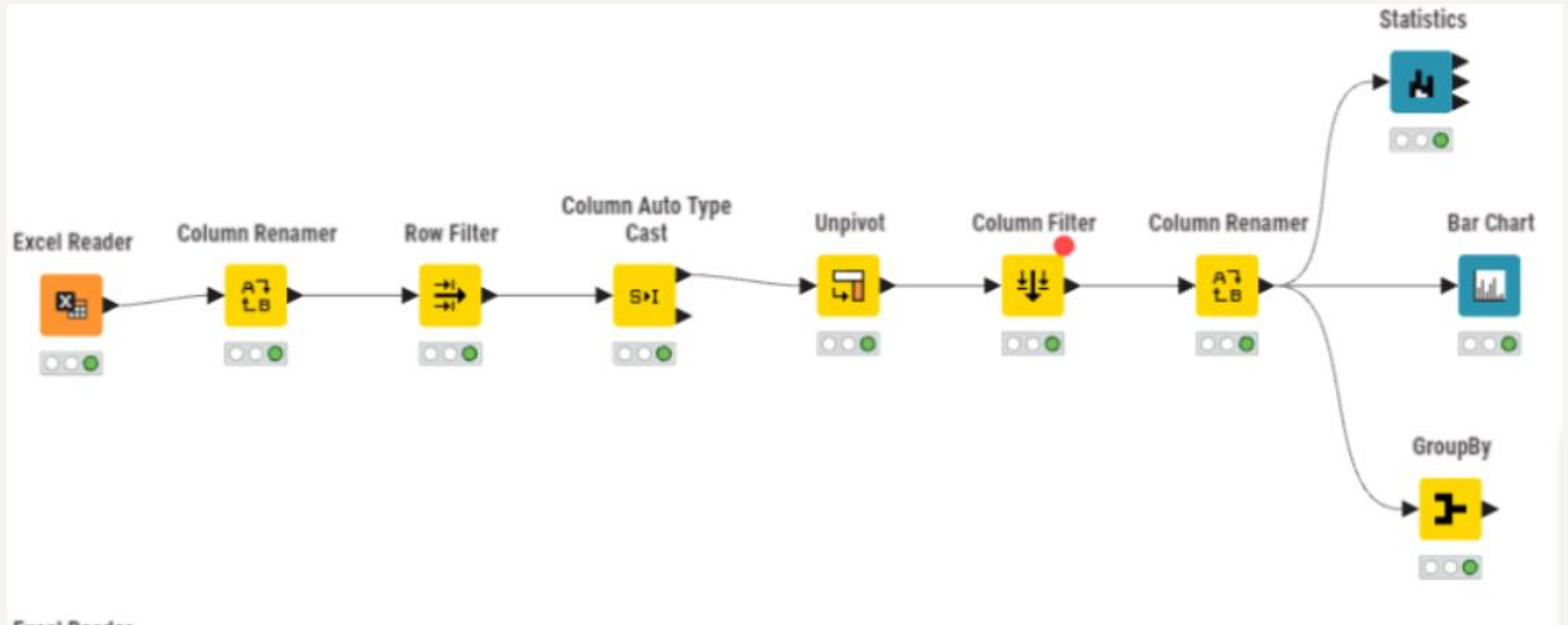
Data Quality Issues

Limitation	What It Means
"n.p." values	Counts under 5 are hidden → some breakdowns incomplete
Vic/NSW coding changes	2012 & 2017 breaks in series → cautious on state trends
Counts only	No population-adjusted rates → can't compare group sizes fairly

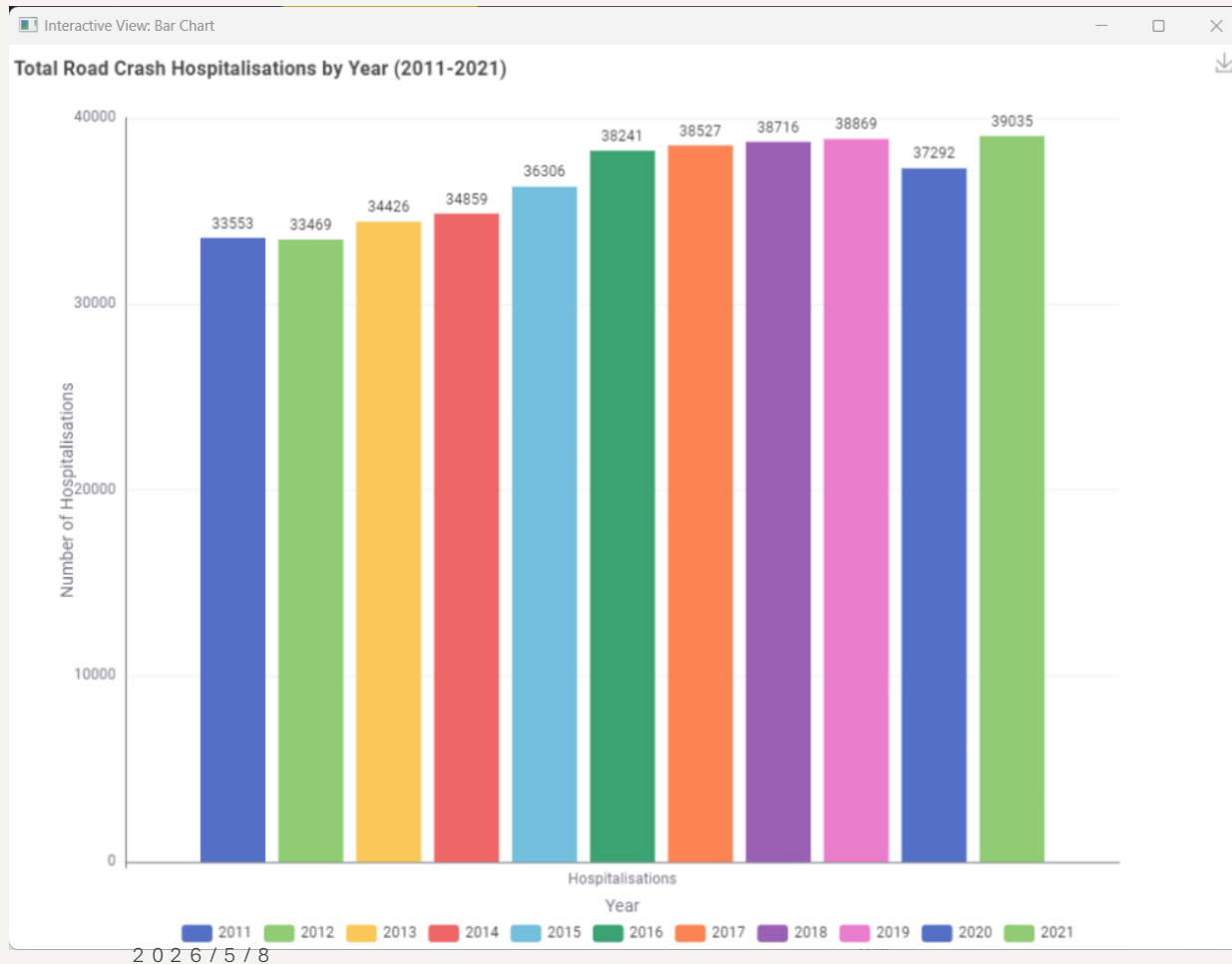
Ethical Considerations

- **We follow ethical guidelines for reporting Indigenous health data**
- **Focus on systemic patterns, not individual behaviour**
- **No personal or identifiable information in the dataset**

Data Processing – KNIME Workflow



Preliminary EDA – National Trend (All Australians)



- A key figure: Total range ~ 33,553 (2011) → 39,035 (2021)
- A line of Observation: Slight dip in 2020, possible COVID lockdown effect

Next Steps

- Apply the same KNIME workflow to the "First-Nations-hospitalised-injuries-2023.xlsx" dataset (Tables 1-4: annual, road user, age group, remoteness)
- Extract the comparison numbers needed for D3.js (First Nations vs Non-Indigenous by each dimension)
- Export cleaned data as CSV for D3 development

Work In Progress

- Refining KNIME to combine all tables (age, road user, remoteness)
- Finalising visualisation design (charts, colours, interactions)
- Preparing D3.js setup for interactive web visualisation.

Q & A

Thank you!

Any Questions?